

- number of samples $n \uparrow \rightarrow$ distribution of the sample mean after normalization converges to a Gaussian distribution
 - ■ $\bar{X} \approx \mathcal{N}(\mu, \sigma^2/n)$ for large n
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Linear Regression

- models the relationship between input features and a target variable
 - assumes a linear relationship between inputs and output
- model
 - $y = Xw + \epsilon$
 - X : input features
 - w : weight vector
 - ϵ : noise term (usually assumed Gaussian)
- objective
 - $\mathcal{L}(w) = \frac{1}{N} \sum_{i=1}^N (y_i - x_i^\top w)^2$
 - minimize prediction error between model output and ground-truth targets
 - typically using mean squared error (MSE)
- interpretation
 - each weight w_j represents the contribution of feature j to the output
 - learns a hyperplane in feature space
 - gradient-based optimization
 - compute gradient of MSE w.r.t. w
 - $\nabla_w \mathcal{L} = -\frac{2}{N} X^\top (y - Xw)$
 - update using gradient descent
 - $w_{t+1} = w_t - \eta \nabla_w \mathcal{L}(w_t)$

- closed-form solution (normal equation)
 - $w^* = (X^\top X)^{-1} X^\top y$
 - derived by setting gradient to zero
 - but $X^\top X$ may be invertible (numerical instability)
 - in practice, gradient descent, SVD etc.
- probabilistic interpretation
 - linear regression with MSE = MLE under Gaussian noise
 - assume Gaussian noise: $\epsilon \sim \mathcal{N}(0, \sigma^2)$
 - target: a linear function of the input plus additive noise
 - likelihood: $p(y|X, w) = \mathcal{N}(Xw, \sigma^2 I)$
 - maximizing log likelihood = minimizing MSE
 - $\log p(y|X, w) = -\frac{1}{2\sigma^2} \|y - Xw\|^2 + \text{const}$
 - L2 regularization $\rightarrow$ MAP with Gaussian prior
 - $p(w) = \mathcal{N}(0, \tau^2 I)$
- limitations
 - cannot model nonlinear relationships
 - unless features are manually engineered
 - sensitive to outliers (due to squared error)

Logistic Regression

$$p(y = 1|x) = \sigma(w^\top x), \sigma(a) = \frac{1}{1+e^{-a}}$$

- linear classifier that models the probability of a binary label
 - uses a linear score $w^\top x$ and squashes it into $[0, 1]$ with a sigmoid
 - decision boundary: predict class 1 when $p(y = 1|x) \geq 0.5$
 - equivalent to $w^\top x \geq 0$
- binary classification model

- $p(y = 1|x) = \sigma(w^\top x), \sigma(a) = \frac{1}{1+e^{-a}}$
 - likelihood (Bernoulli)
 - $p(y|x, w) = p^y(1 - p)^{1-y}$ where $p = \sigma(w^\top x)$
 - $p(y|x, w) = \sigma(w^\top x)^y(1 - \sigma(w^\top x))^{1-y}$
 - MLE objective

$$p = \sigma(w^\top x) = \frac{1}{1 + e^{-w^\top x}}$$

 - maximize likelihood over data
 - equivalent to minimizing NLL (negative log-likelihood)
 - NLL = binary cross-entropy
 - $-\log p(y|x, w) = -y \log p - (1 - y) \log(1 - p)$
 - equivalent to Binary Cross Entropy (BCE)
 - gradient: $\nabla_w \mathcal{L}(w) = X^\top (p - y)$
 - training
 - no closed-form solution in general
 - optimize with gradient descent or variants (SGD, Adam)
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Convexity

any local minimum = global minimum

- determines whether a problem has a single global minimum
 - a property of functions and optimization problems
- intuition
 - a function is convex if the line segment between any two points lies above the function
 - no local minima other than the global minimum

Convex Function

- definition